A FINITE TYPE ALGEBRA OVER A HENSELIAN PAIR THAT IS QUASI-FINITE OVER THE CLOSED FIBRE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Henselian pair and the lifting of idempotents along it. the statement counted by the sibling card S1-algebra-lift-idempotent at 1784-1803. Supplies the decomposition of the algebra into a product of rings.
- A finite type algebra over a henselian pair whose closed fibre is finite over the residue field splits into a product over the points of the fibre. used at proof lines 2512-2540. Provides the factorisation of the algebra.
- Quasi-finiteness of an algebra over a pair and its behaviour under the decomposition. used at proof lines 2540-2550. Supplies the property that the pieces have to satisfy.

Source reference. Original lemma 09XH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12530–12530. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00PL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 935–935. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00Q9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 959–959. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00HZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 661–661. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00GU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 620–620. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00EM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 540–540. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00HN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 649–649. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let (A, I) be a pair. Let $A \to B$ be a finite type ring map such that $B/IB = C_1 \times C_2$ with $A/I \to C_1$ finite. Let B' be the integral closure of A in B. Then we can write $B'/IB' = C_1 \times C_2'$ such that the map $B'/IB' \to B/IB$ preserves product decompositions and there exists a $g \in B'$ mapping to $(1, 0)$ in $C_1 \times C_2'$ with $B'_g \to B_g$ an isomorphism.*

Proof. Observe that $A \to B$ is quasi-finite at every prime of the closed subset $T = \mathrm{Spec}(C_1) \subset \mathrm{Spec}(B)$ (this follows by looking at fibre rings, see Algebra, Definition 00PL). Consider the diagram of topological spaces

$$\begin{array}{ccc} \mathrm{Spec}(B) & \xrightarrow{\quad\phi\quad} & \mathrm{Spec}(B') \\ & {\scriptstyle\psi}\searrow \quad \swarrow{\scriptstyle\psi'} & \\ & \mathrm{Spec}(A) & \end{array}$$

By Algebra, Theorem 00Q9 for every $\mathfrak{p} \in T$ there is a $h_\mathfrak{p} \in B'$, $h_\mathfrak{p} \notin \mathfrak{p}$ such that $B'_h \to B_h$ is an isomorphism. The union $U = \bigcup D(h_\mathfrak{p})$ gives an open $U \subset \mathrm{Spec}(B')$ such that $\phi^{-1}(U) \to U$ is a homeomorphism and $T \subset \phi^{-1}(U)$. Since T is open in $\psi^{-1}(V(I))$ we conclude that $\phi(T)$ is open in $U \cap (\psi')^{-1}(V(I))$. Thus $\phi(T)$ is open in $(\psi')^{-1}(V(I))$. On the other hand, since C_1 is finite over A/I it is finite over B'. Hence $\phi(T)$ is a closed subset of $\mathrm{Spec}(B')$ by Algebra, Lemmas 00HZ and 00GU. We conclude that $\mathrm{Spec}(B'/IB') \supset \phi(T)$ is open and closed. By Algebra, Lemma 00EM we get a corresponding product decomposition $B'/IB' = C_1' \times C_2'$. The map $B'/IB' \to B/IB$ maps C_1' into C_1 and C_2' into C_2 as one sees by looking at what happens on spectra (hint: the inverse image of $\phi(T)$ is exactly T; some details omitted). Pick a $g \in B'$ mapping to $(1, 0)$ in $C_1' \times C_2'$ such that $D(g) \subset U$; this is possible because $\mathrm{Spec}(C_1')$ and $\mathrm{Spec}(C_2')$ are disjoint and closed in $\mathrm{Spec}(B')$ and $\mathrm{Spec}(C_1')$ is contained in U. Then $B'_g \to B_g$ defines a homeomorphism on spectra and an isomorphism on local rings (by our choice of U above). Hence it is an isomorphism, as follows for example from Algebra, Lemma 00HN. Finally, it follows that $C_1' = C_1$ and the proof is complete. □